

Use-case flow

Use case: Monitor service health & dispatch downtime alert

Actors: System (automated scheduler), DevOps Engineer (notified), Notification Service (external, delivers the alert)

Preconditions:

- Microservice is registered in the catalog with a valid health endpoint (FR-002).
- An alert rule is configured for the service, or the system default rule applies (FR-005).

Postconditions:

- *Success:* rolling 24-hour availability is updated; if a breach occurred, an alert record is created and dispatched.
- *Failure:* the probe result is still logged even if the notification channel is unreachable — no data is silently lost.

Main success scenario (Steps):

1. Scheduler triggers a health probe for the service at the 30-second interval.
2. System sends an HTTP request to the service's health endpoint.
3. Service responds with 200 OK within the timeout window.
4. System records the probe as "healthy" with a timestamp.
5. System recomputes the rolling 24-hour availability percentage.
6. Updated percentage is reflected on the DevOps Engineer's dashboard in real time.

Alternate flow 3a - three consecutive failed probes:

- 3a1. At step 3, the service fails to respond (timeout, non-2xx, or connection refused).
- 3a2. System records the probe as "failed" and increments the consecutive-failure counter.
- 3a3. Steps 1–2 repeat for the next two scheduled probes.
- 3a4. On the 3rd consecutive failure, the extend point triggers *Dispatch downtime alert*.
- 3a5. System looks up the configured alert rule and sends the alert to the Notification Service.
- 3a6. Flow resumes at main step 6 with availability reflecting the downtime.

Alternate flow 5a - notification channel unreachable:

- 5a1. Notification Service fails to deliver the alert (e.g., Slack webhook timeout).

- 5a2. System retries the dispatch once, after a 60-second delay.
- 5a3. If the retry also fails, the alert is logged as "undelivered" and escalated to the default email channel.
- 5a4. Flow resumes at main step 6.